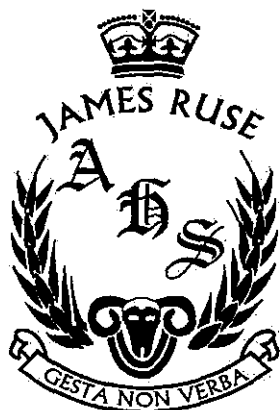


Student Number:	
Class:	



TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION 2015

MATHEMATICS EXTENSION 2

General Instructions:

- Reading Time: 5 minutes.
- Working Time: 3 hours.
- Write in black pen.
- Board approved calculators & templates may be used
- A Standard Integral Sheet is provided.
- In Question 11 - 16, show all relevant mathematical reasoning and/or calculations.
- Marks may not be awarded for careless or badly arranged working.

Total Marks 100

Section I: 10 marks

- Attempt Question 1 – 10.
- Answer on the Multiple Choice answer sheet provided.
- Allow about 15 minutes for this section.

Section II: 90 Marks

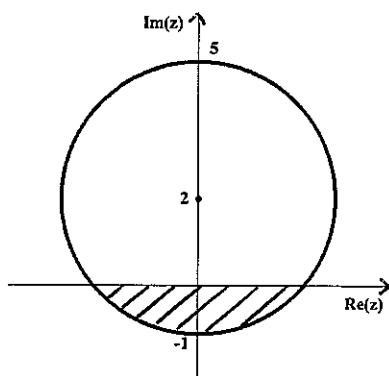
- Attempt Question 11 - 16
- Answer on lined paper provided. Start a new page for each new question.
- Allow about 2 hours & 45 minutes for this section.

The answers to all questions are to be returned in separate *stapled* bundles clearly labelled Question 11, Question 12, etc. Each question must show your Candidate Number.

Section 1 (10 Marks)

Attempt questions 1 – 10. Use the Multiple Choice answer sheet supplied.

1. A circle with centre $(0,2)$ and radius 3 units is shown below on an Argand diagram.



Which of the following inequalities represents the shaded region ?

- A) $\text{Im}(z) \leq 0$ and $|z - 2| \leq 3$ B) $\text{Re}(z) \leq 0$ and $|z - 2| \leq 3$
C) $\text{Im}(z) \leq 0$ and $|z - 2i| \leq 3$ D) $\text{Re}(z) \leq 0$ and $|z - 2i| \leq 3$
2. The equation $x^2 - xy + y^2 = 3$ defines y implicitly in terms of x .
The expression for $\frac{dy}{dx}$ is :
- A) $\frac{y-2x+3}{2y-x}$ B) $\frac{y-2x+3}{x-2y}$ C) $\frac{y-2x}{2y-x}$ D) $\frac{y-2x}{x-2y}$

3. In how many ways can five letters be chosen from the letters ARRANGE ?

A) 9 B) 10 C) 12 D) 21

4. Consider a polynomial $P(x)$ of degree 3.

Two real numbers a and b are such that :

$$a < b$$

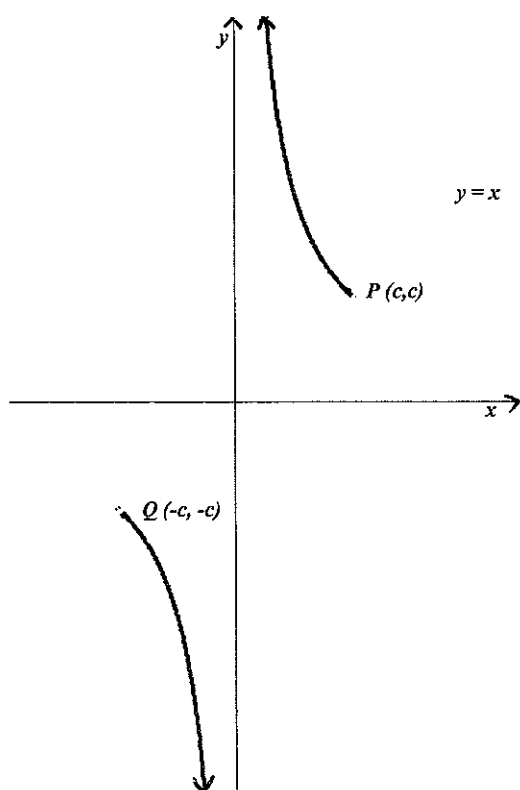
$$P(a) > P(b) > 0$$

$$P'(a) = P'(b) = 0$$

The polynomial has :

- A) 3 real zeroes B) 1 real zero γ such that $\gamma < a$
C) 1 real zero γ such that $a < \gamma < b$ D) 1 real zero γ such that $\gamma > b$

5. The graph shows a part of the hyperbola $x = ct, y = c/t$.

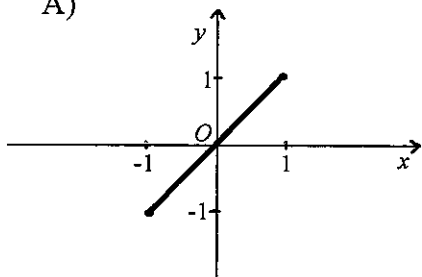


Which pair of parametric equations precisely describes the graph as shown?

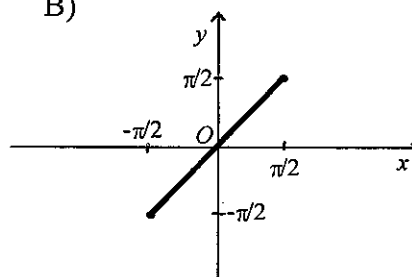
- A) $x = c(t^2 + 1), y = c/(t^2 + 1)$ B) $x = c(1 - t^2), y = c/(1 - t^2)$
 C) $x = c\sqrt{1 - t^2}, y = c/\sqrt{1 - t^2}$ D) $x = c \sin t, y = c/\sin t$
6. The region enclosed by $y = x^3, y = 0$ and $x = 2$ is rotated about the y -axis to produce a solid. What is the volume of that solid?
- A) $\frac{8\pi}{5} \text{ units}^3$ B) $\frac{32\pi}{5} \text{ units}^3$ C) $\frac{64\pi}{5} \text{ units}^3$ D) $\frac{16\pi}{5} \text{ units}^3$
7. The equation $|z - 4| + |z + 4| = 10$ defines an ellipse. What is the length of the semi minor axis?
- A) $2\frac{2}{5}$ B) 3 C) 4 D) 5
8. A particle is moving in a circle of radius 80cm with a linear speed of 4π m/s. It has a constant angular speed (in revolutions per minute) of:
- A) $3/8$ rpm B) $3/2$ rpm C) $37\frac{1}{2}$ rpm D) 150 rpm

9. Which of the diagrams below best represents the graph of $y = \sin^{-1}(\sin x)$?

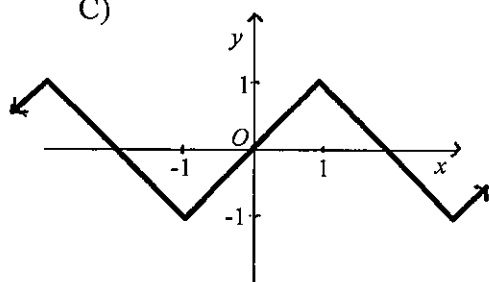
A)



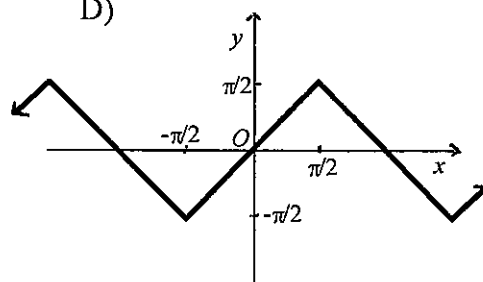
B)



C)



D)



10. A particle of mass m moves in a straight line under the action of a resultant force F where $F=F(x)$. Given that the velocity v is v_0 when the position x is x_0 , and that v is v_1 when x is x_1 , it follows that $|v_1| =$

- A) $\sqrt{\frac{2}{m} \int_{x_0}^{x_1} \sqrt{F(x)} dx} + v_0$
- B) $\sqrt{2} \int_{\sqrt{x_0}}^{\sqrt{x_1}} F(x) dx + v_0$
- C) $\sqrt{\frac{2}{m} \int_{x_0}^{x_1} \sqrt{F(x)} dx + (v_0)^2}$
- D) $\sqrt{\frac{2}{m} \int_{x_0}^{x_1} \{F(x) + (v_0)^2\} dx}$

End of Section 1

Section 2 (90 Marks)

Attempt all questions 11 – 16. Start each question on a new sheet of paper.

QUESTION 11 (Start a new sheet of paper)

Marks

a) Find $\int \frac{1}{\sqrt{5+4x-x^2}} dx$

2

b) i) Find the real numbers a and b such that

$$\frac{3x^2-3x+7}{(x-2)(x^2+9)} \equiv \frac{a}{(x-2)} + \frac{bx+1}{(x^2+9)}$$

2

ii) Find $\int \frac{3x^2-3x+7}{(x-2)(x^2+9)} dx$

3

c) i) Prove that $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

1

ii) Hence find the value of $\int_0^2 x(2-x)^5 dx$.

2

d) Consider the equation $z^2 + az + (2+i) = 0$.

i) Find the complex number a , given that i is a root of the equation.

1

ii) Also write down the second root of the equation.

1

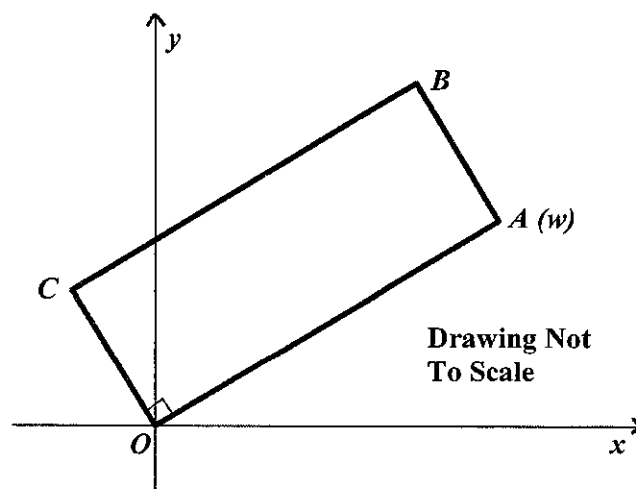
e) In the Argand diagram, $OABC$ is a rectangle, where $3OC = OA$. The vertex at A represents the complex number ω .

i) What complex number corresponds to the vertex C ?

1

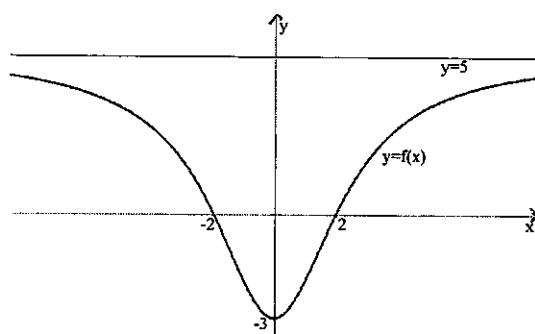
ii) What complex number corresponds to the point of intersection D of the diagonals OB and AC ?

2



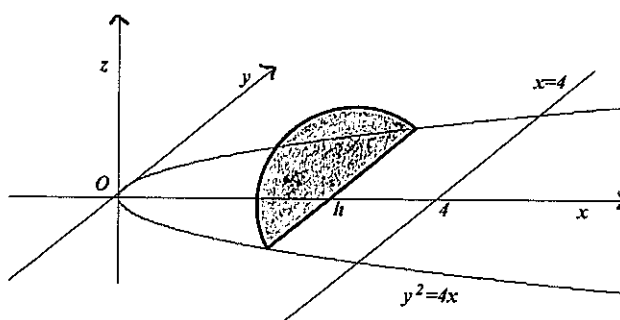
QUESTION 12 (Start a new sheet of paper)

- a) The diagram below shows the graph of $y = f(x)$.



Draw separate one-third page sketches of the following graphs:

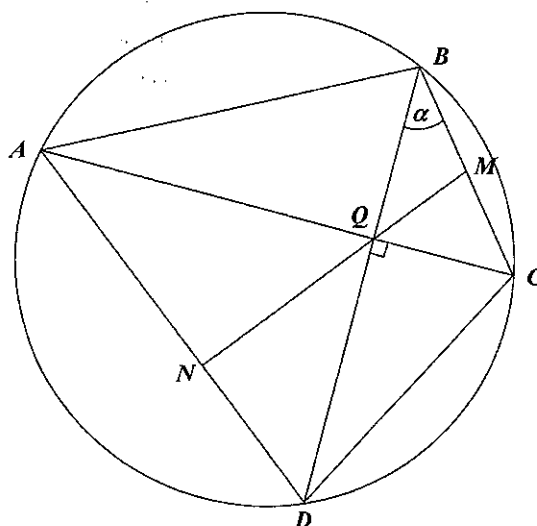
- | | | |
|------|----------------------|---|
| i) | $y = \frac{1}{f(x)}$ | 2 |
| ii) | $y^2 = f(x)$ | 2 |
| iii) | $y = (f(x))^2$ | 2 |
- b) i) The polynomial $P(x) = x^4 - 6x^3 + 13x^2 - ax - b$ has two double zeroes. Find a and b . 3
- ii) Hence determine the equation of the line which touches the curve $y = x^4 - 6x^3 + 13x^2$ at two distinct points. 1
- c) The base of a solid S is the region in the xy plane enclosed by the parabola $y^2 = 4x$ and the line $x = 4$. Each cross section perpendicular to the x axis is a semi-ellipse with the major axis in the xy plane and with the major and minor axes in the ratio $a:b$.



- | | | |
|------|---|---|
| i) | Assuming that the area of an ellipse with semi-axes A and B is πAB , show that the area of the semi-ellipse shown at $x = h$ is $2\pi hb/a$. | 1 |
| ii) | Find the volume of the solid S . | 3 |
| iii) | The solid T is obtained by rotating the region enclosed by the parabola and the line $x = 4$ about the x axis. Using (with a little care) your result from (ii), or otherwise, find the volume of T . | 1 |

QUESTION 13 (Start a new sheet of paper)

a)

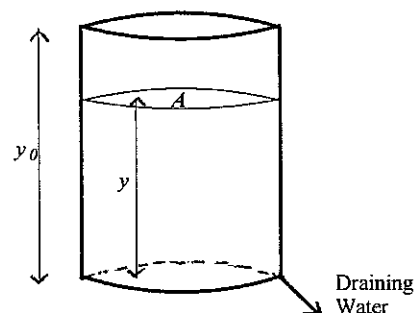


In the diagram above $ABCD$ is a cyclic quadrilateral whose diagonals are perpendicular and intersect at Q . Let M be the midpoint of BC and suppose that MQ produced meets AD at N . Let $\angle QBC = \alpha$.

- i) Explain why $BM = QM$. 1
- ii) Prove that $MN \perp AD$. 3

- b) At a point on a railway line where the radius of curvature is 200m, the track is designed so that a train travelling at 50 km/hr exerts no lateral force on the track. What lateral force would a stationary locomotive, of mass 40 tonnes, exert on the track at this point? (Take g to be 9.81m/s.s.) 4

- c) The diagram shows the side view of a vertical cylindrical water cooler of constant cross sectional area A . Water drains through a hole at the bottom of the cooler. It is known that the volume of water decreases at a rate given by $\frac{dv}{dt} = -k\sqrt{y}$ where k is a positive constant and y is the depth of the water. Initially the cooler is full and it would take T seconds to drain completely.



- i) Show that $\frac{dy}{dt} = -\frac{k}{A}\sqrt{y}$ 1
- ii) Show that $y = y_0 \left(1 - \frac{t}{T}\right)^2$ for $0 \leq t \leq T$. 4
- iii) If it takes 10 seconds for half the water to drain, evaluate T . 2

QUESTION 14 (Start a new sheet of paper)

- a) A particle moves in Simple Harmonic Motion, the period being 2 seconds and the amplitude 3 metres. Find the maximum speed and the maximum acceleration during the motion. 2

- b) i) Use de Moivre's Theorem to show that

$$(\cot\theta + i)^n + (\cot\theta - i)^n = \frac{2\cos n\theta}{\sin^n\theta} \quad 2$$

- ii) Show that the equation $(x + i)^5 + (x - i)^5 = 0$ has roots $0, \pm\cot\frac{\pi}{10}, \pm\cot\frac{3\pi}{10}$. 2

- iii) Hence show that the equation $x^4 - 10x^2 + 5 = 0$ has roots $\pm\cot\frac{\pi}{10}, \pm\cot\frac{3\pi}{10}$. 2

- iv) Hence show that $\cot\frac{\pi}{10} = \sqrt{5 + 2\sqrt{5}}$ 2

- c) A ball is projected so as to just clear two walls. The first wall is at a height b at a horizontal distance a from the point of projection and the second is of height a at a horizontal distance b from the point of projection. ($b > a$)

It may be assumed that, if the ball is projected from the origin with velocity V at an angle α to the horizontal (x axis), then the equation of the path is given by

$$y = x\tan\alpha - \frac{gx^2\sec^2\alpha}{2V^2}$$

- i) Show that the range on the horizontal plane is $\frac{a^2+ab+b^2}{a+b}$. 3
- ii) Show that the angle of projection must exceed $\tan^{-1}3$. 2

QUESTION 15 (Start a new sheet of paper)

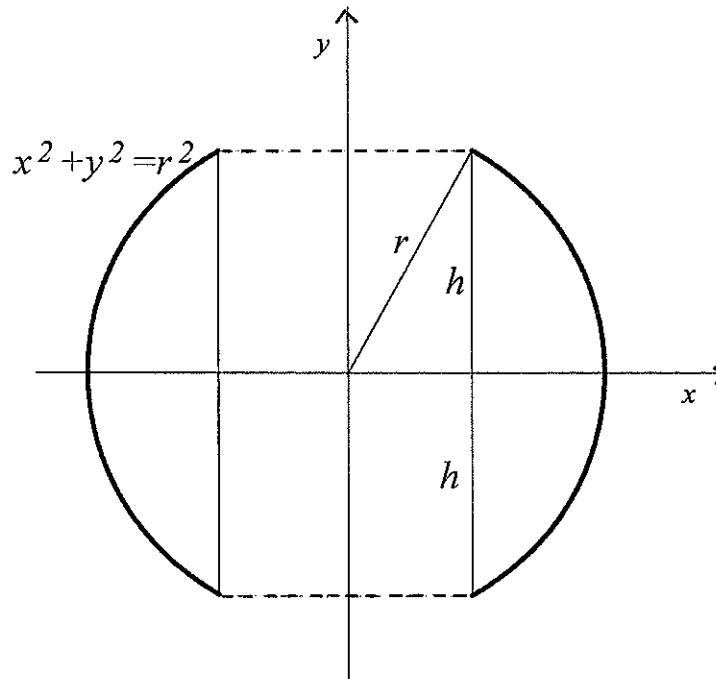
- a) The diagram shows a segment of the circle $x^2 + y^2 = r^2$ which is rotated about the y axis to form a collar. This collar is thus a sphere with a symmetrical hole through it. Let the hole be of height $2h$ as shown.

Use the method of cylindrical shells to show that the volume of the material in the collar is given by the integral

$$4\pi \int_{\sqrt{r^2-h^2}}^r x\sqrt{r^2-x^2} dx$$

Evaluate the integral to show that the volume of material in the collar is a function of h only and independent of r .

4

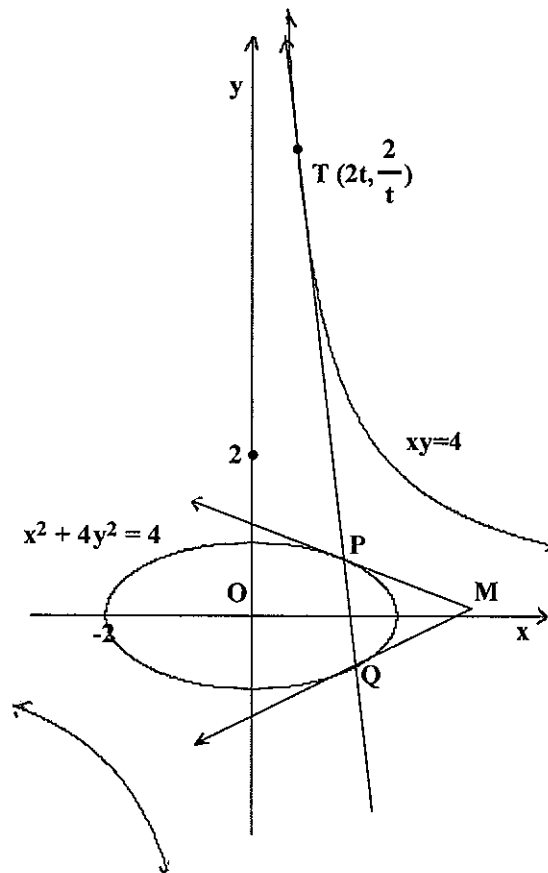


- b) Consider the equation $z^5 + 32 = 0$.
- Write down the roots of this equation in modulus argument form. 2
 - Illustrate these roots on an Argand diagram. 1
 - If the points A, B, C, D and E cyclically represent these roots, find the area of triangle ACD . (Give your answer to 2 decimal places) 3

Question 15 is continued on the next page

Question 15 (continued)

- c) i) Consider the diagram below. The tangent at $T(2t, \frac{2}{t})$ to the rectangular hyperbola $xy = 4$ meets the ellipse $x^2 + 4y^2 = 4$ at P and Q . The tangents to the ellipse at P and Q intersect at M . Find the equation of the locus of M . (You may use standard forms of tangents and such things without proof.) 3
- ii) Describe briefly the locus and any restrictions that it may have. 2



QUESTION 16 (Start a new sheet of paper)

- a) A box contains n jellybeans, some white and some black. Alan and Betty take turns picking a jellybean from the box, without looking, until the box is empty. Alan picks first.
- i) If there is 1 black and $n - 1$ white jellybeans and n is odd, find the probability that Alan picks the black jellybean. 1
- ii) If there are 2 black and $n - 2$ white jellybeans and n is even, find the probability that Alan is the first to pick a black jellybean. 2
- iii) If there are 2 black and $n - 2$ white jellybeans and n is odd, find the probability that Alan is the first to pick a black jellybean. 2

Question 16 is continued on the next page....

Question 16 (continued)

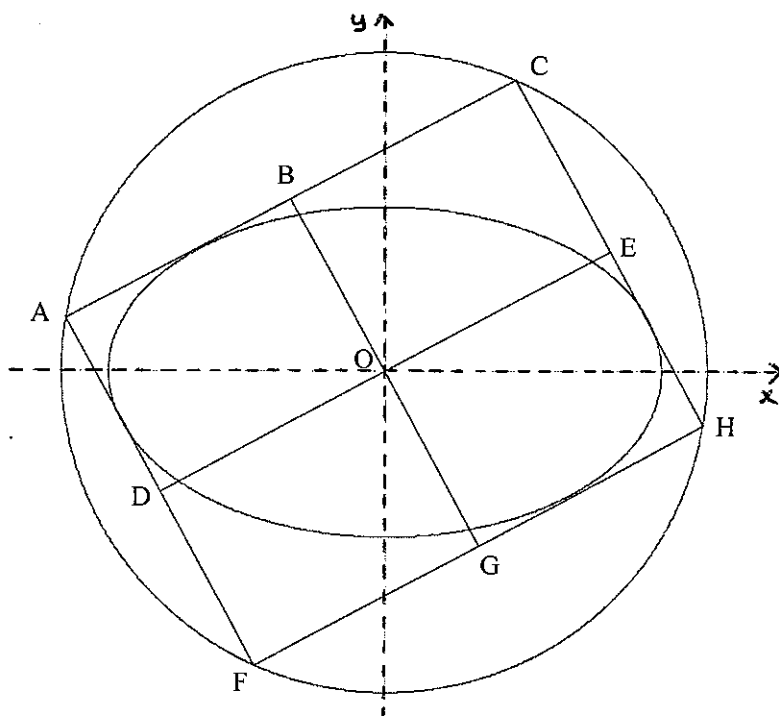
b) Consider the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

i) Show that lines with equations $y = mx \pm \sqrt{a^2m^2 + b^2}$ are always tangent to the ellipse for any real value of m . 2

ii) From any point external to the ellipse, two tangents may be drawn. By considering the above as a quadratic equation in m , or otherwise, show that the locus of points where the two tangents are perpendicular to each other is a circle with equation $x^2 + y^2 = a^2 + b^2$. 2

iii) Show that the area of the ellipse's circumscribing rectangle of which $y = mx \pm \sqrt{a^2m^2 + b^2}$ are two parallel sides, is given by

$$A = \frac{4}{(1+m^2)} \sqrt{(a^2 + m^2b^2)(a^2m^2 + b^2)} \quad \text{3}$$



You are given that this formula for the area can be rearranged to give the following form for the square of the area: $A^2 = 16a^2b^2 + \frac{16(a^2 - b^2)^2}{(m + \frac{1}{m})^2}$

iv) Show that, for any $m > 0$, $m + \frac{1}{m} \geq 2$. 1

v) Hence, or otherwise, find the maximum and minimum areas of rectangles which circumscribe the ellipse. 2

END OF EXAM

JRAHS EXTENSION 2 MATHS TRIAL 2015 (SOLUTIONS)

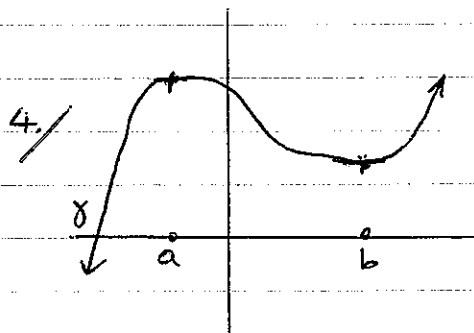
2/ $2x - y - x \frac{dy}{dx} + 2y \frac{dy}{dx} = 0$

$$\frac{dy}{dx}(2y - x) = y - 2x$$

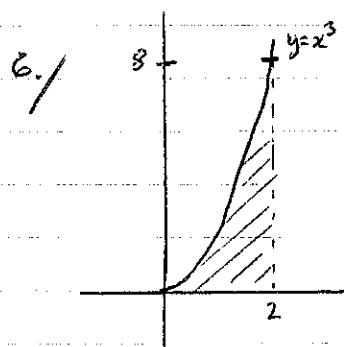
$$\frac{dy}{dx} = \frac{y - 2x}{2y - x}$$

3/

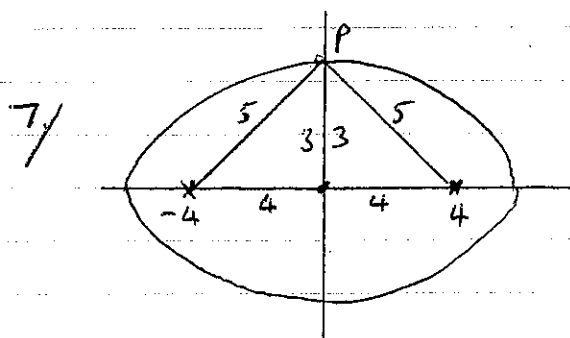
2R	2A	3 ways
1R	2A	3 "
2R	1A	3 "
1R	1A	1 "
0R	2A	1 "
2R	0A	1 "
		<u>12 ways</u>



- 1/ B
- 2/ C
- 3/ C
- 4/ B
- 5/ D
- 6/ C
- 7/ B
- 8/ D
- 9/ D
- 10/ C



$$\begin{aligned} & \pi \cdot 2^2 \cdot 8 - \int_0^8 \pi x^2 dy \\ &= \pi \left(32 - \int_0^8 y^{2/3} dy \right) \\ &= \pi \left(32 - \left[\frac{3y^{5/3}}{5} \right]_0^8 \right) \\ &= \pi \left(32 - 32 \times \frac{3}{5} \right) \\ &= \underline{64\pi/5} \end{aligned}$$



8/ $0.8 \omega = 4\pi$ (m/sec)
 $\omega = 4\pi/0.8 = 5\pi$ (rad/sec)
 $= 300\pi$ rad/min
 $= \underline{150 \text{ rev/min}}$

10/

$$\begin{aligned} F(x) &= m \ddot{x} \\ &= m \frac{d}{dx} \left(\frac{v^2}{2} \right) \\ \int_{x_0}^{x_1} F(x) dx &= m \left[\frac{v^2}{2} \right]_{v_0}^{v_1} \\ \frac{2}{m} \int_{x_0}^{x_1} F(x) dx &= v_1^2 - v_0^2 \\ v_1^2 &= \frac{2}{m} \int_{x_0}^{x_1} F(x) dx + v_0^2 \\ |v_1| &= \sqrt{\frac{2}{m} \int_{x_0}^{x_1} F(x) dx + v_0^2} \end{aligned}$$

Question 11

$$a) \int \frac{dx}{\sqrt{9-(x-2)^2}} = \frac{\sin^{-1}(\frac{x-2}{3})}{1} + k$$

$$b) i) a(x^2+9) + (bx+1)(x-2) \\ \equiv 3x^2 - 3x + 7$$

Set $x=2$

$$13a = 13 \quad \underline{a=1}$$

$$\text{Coeff of } x^2: a+b=3 \therefore \underline{b=2}$$

$$(ii) \int \frac{3x^2-3x+7}{(x-2)(x^2+9)} dx = \int \frac{dx}{(x-2)} + \int \frac{2x+1}{x^2+9} dx$$

$$= \int \frac{1}{(x-2)} dx + \int \frac{2x}{x^2+9} dx + \int \frac{1}{x^2+9} dx$$

$$= \ln|x-2| + \ln(x^2+9) + \frac{1}{3} \tan^{-1}\left(\frac{x}{3}\right) + k$$

$$c) \text{ Let } I = \int_0^a f(x) dx$$

$$\text{Set } u = a-x \rightarrow x = a-u \\ "du = -dx"$$

$$\text{When } x=a, u=0 \\ x=0, u=a$$

$$\therefore I = \int_a^0 f(a-u) \cdot -du$$

$$= - \int_a^0 f(a-u) du$$

$$= \int_0^a f(a-u) du$$

$$= \int_0^a f(a-x) dx$$

(Dummy variable)

$$ii) I = \int_0^2 x(2-x)^5 dx$$

$$= \int_0^2 (2-x)x^5 dx \quad (\text{Using (i)})$$

$$= \int_0^2 2x^5 - x^6 dx$$

$$= \left[\frac{x^6}{3} - \frac{x^7}{7} \right]_0^2 = \frac{64}{3} - \frac{128}{7}$$

$$= \frac{7 \times 64 - 3 \times 128}{21} = \underline{\underline{\frac{64}{21}}}$$

d) i) By factor thm:

$$(i)^2 + a(i) + 2 + i = 0$$

$$i(a+1) = -1$$

$$a+1 = -1/i = i$$

$$\underline{a = i-1}$$

ii) Let other root be β

$$\text{Sum of roots} = -a = \beta + i$$

$$\therefore (1-i) = \beta + i$$

$$\beta = (1-i) - i$$

$$= \underline{\underline{1-2i}}$$

e) i) Rotation $\equiv$ Multiply by i
The length is shortened by a factor of 3.

$$\therefore C \text{ represents } \underline{\underline{i\omega/3}}$$

ii) Diagonals bisect each other.

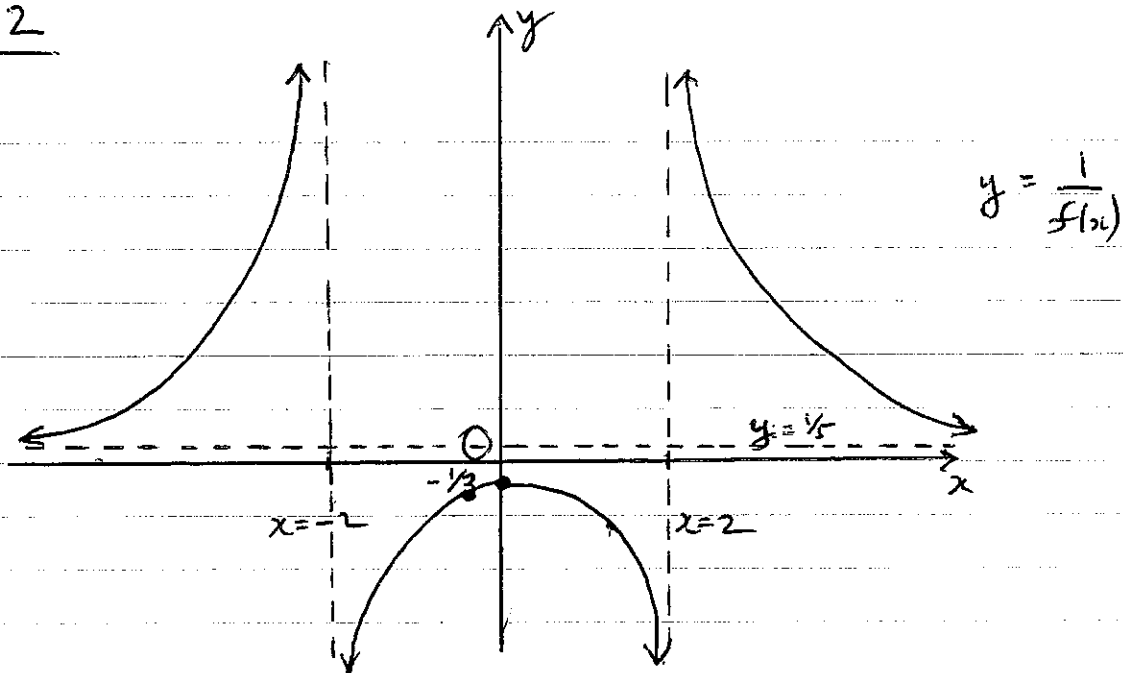
$$|OD| = \frac{1}{2}|OB|$$

$$OB = OA + AB = OA + OC$$

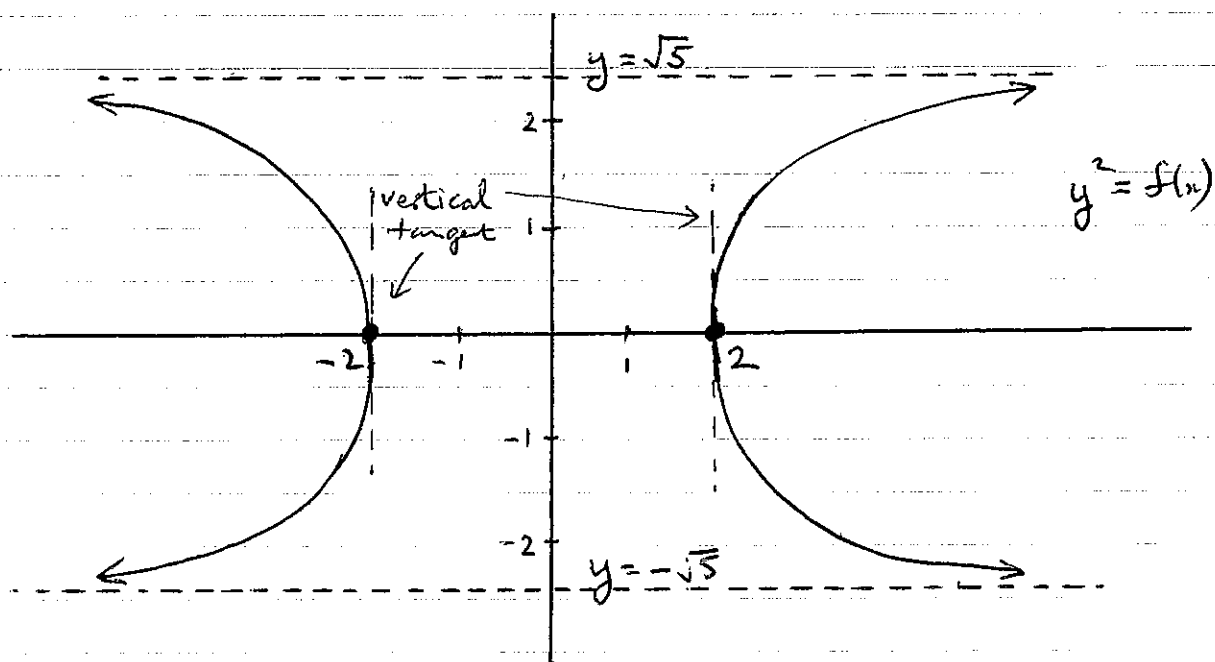
$$OD = \frac{1}{2} \left(\omega + \frac{i\omega}{3} \right) = \underline{\underline{\omega \left(\frac{1}{2} + \frac{i}{6} \right)}}$$

Question 12

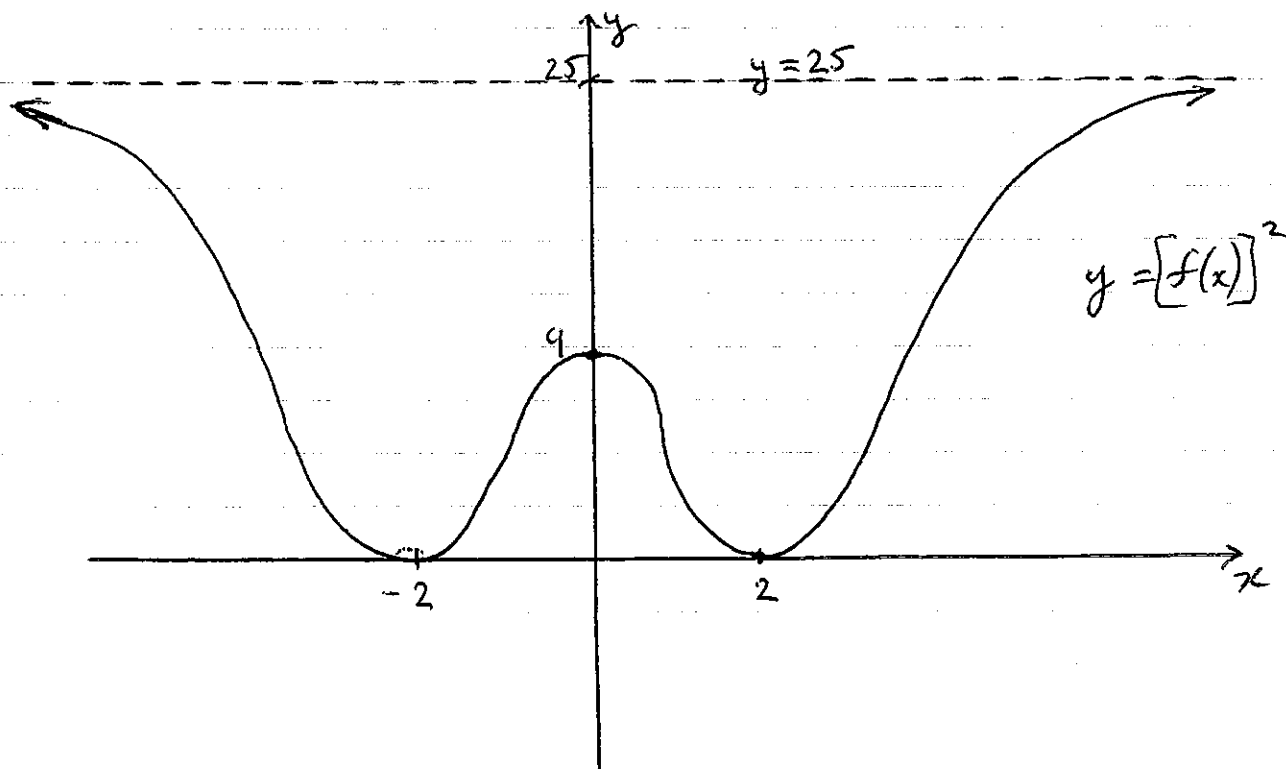
a) i)



ii)



iii)



Question 12 (cont)

b) i) Let the zeros of $P(x)$ be $\alpha, \alpha, \beta, \beta$.

$$\therefore 2(\alpha + \beta) = 6 \quad (\text{sum of roots})$$

$$\underline{\alpha + \beta = 3} \quad (1)$$

$$\alpha^2 + \beta^2 + 4\alpha\beta = 13 \quad (\text{sum of roots 2 at a time})$$

$$(\alpha + \beta)^2 + 2\alpha\beta = 13$$

$$2\alpha\beta = 4 \quad (\text{Using (1) above})$$

$$\underline{\alpha\beta = 2}$$

Now $b = -\alpha^2\beta^2$ (Product of roots) $a = 2\alpha^2\beta + 2\beta^2\alpha$ (Sum of roots 3 at a time)

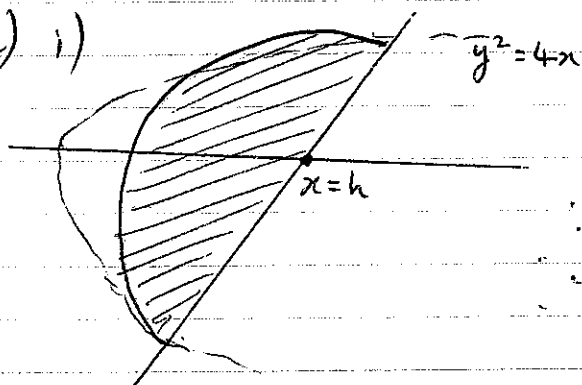
$$= -(\alpha\beta)^2 = 2\alpha\beta(\alpha + \beta)$$

$$\underline{b = -4} \quad \underline{a = 12}$$

ii) Two touching points is the above scenario if we solve $x^4 - 6x^3 + 13x^2$ against $ax + b$

$\therefore$ line is $y = 12x - 4$

c) i)



If $x=h$, $y = \pm 2\sqrt{h}$

Length of major axis is $4\sqrt{h}$.

Length of minor axis is $4b\sqrt{h}/a$

Semi-Major, Semi-Minor are $2\sqrt{h}$, $2b\sqrt{h}/a$

Area of Full Ellipse = $4\pi hb/a$

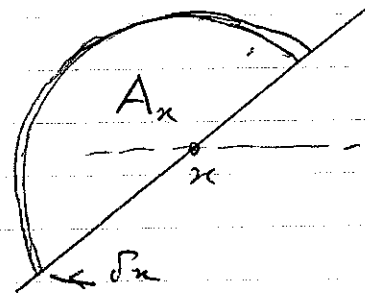
Area of Half Ellipse shown = $2\pi hb/a$

ii) $\delta V = \frac{2\pi x b}{a} \delta x$

$$V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^h \frac{2\pi x b}{a} \delta x$$

$$= \frac{2\pi b}{a} \int_0^h x dx = \frac{2\pi b}{a} \left[\frac{x^2}{2} \right]_0^h$$

$$= \underline{\underline{16\pi b/a \text{ units}^3}}$$

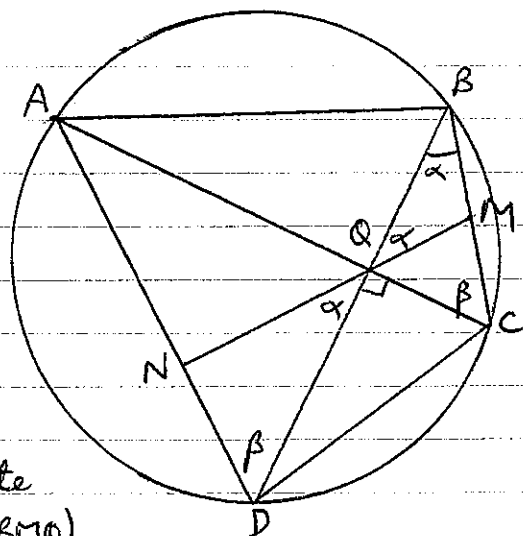


iii) This will generate circles rather than semi ellipses
Hence ratio $b/a = 1$. But it will include two halves

$$\therefore V_{\text{total}} = \underline{\underline{32\pi \text{ units}^3}}$$

Question 13

- a) i) $\angle BQC$ is right angle
(Given that diagonals perpendicular)
 ΔBQC is an angle in a semicircle with BC as diameter and M as the circle centre.
 MB and QM are radii of this circle
Thus equal. i.e. $BM = QM$

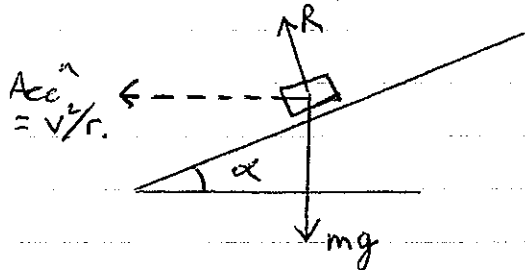


- ii) $\therefore \angle BQM = \alpha$ (Equal angles opposite equal sides in ΔBQM)
 $\therefore \angle NQD = \alpha$ (Vertically opposite angles are equal)

Now let $\angle ACB = \beta$
Then $\angle ADQ = \beta$ (Angles standing on the same arc are equal)

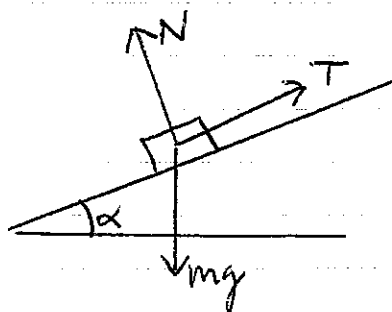
In ΔBQC , $\alpha + \beta + 90^\circ = 180^\circ$
In ΔQND , $\alpha + \beta + \angle QND = 180^\circ$ (Angle sum of triangle is 180°)
 $\therefore \angle QND = 90^\circ$
i.e. $MN \perp AD$

- b) Let angle of rail inclination be α . There is no lateral force at the design speed, 50 km/hr
 $50 \text{ km/h} = \frac{50000}{3600} \text{ m/s} = 13.89 \text{ m/s}$



Resolving vertically $R \cos \alpha = mg$ ①
horizontally $R \sin \alpha = mv^2/r$ ②

② ÷ ① $\tan \alpha = \frac{v^2}{rg} = \frac{(13.89)^2}{200 \times 9.81} = 0.0983$



No movement or acceleration
Resolve vertically $N \cos \alpha + T \sin \alpha = mg$
horizontally $N \sin \alpha - T \cos \alpha = 0$
 $\therefore N = T \cot \alpha$

$\therefore \frac{T \cos^2 \alpha}{\sin \alpha} + T \sin \alpha = mg$
 $T(\cos^2 \alpha + \sin^2 \alpha) = mg \sin \alpha$

$\therefore$ With $m = 40000$, $\sin \alpha = 0.09785$

$T = 38396 \text{ N} = 38.4 \text{ kN}$ (to 3sf)

Question 13 (cont)

c) i) $V = Ay$
 $\therefore \frac{dV}{dt} = A \frac{dy}{dt}$
 $\therefore \frac{dy}{dt} = \frac{1}{A} \frac{dV}{dt} = \frac{1}{A} (-k\sqrt{y}) = \underline{\underline{-\frac{k\sqrt{y}}{A}}}$

ii) $\therefore \frac{dt}{dy} = \frac{-A}{k} y^{-1/2}$

$$t = -\frac{2A}{k} y^{1/2} + c$$

When $t=T, y=0 \therefore c=T$

$$\therefore t = -\frac{2A\sqrt{y}}{k} + T \quad *$$

When $t=0, y=y_0$

$$\therefore T = \frac{2A\sqrt{y_0}}{k} \Rightarrow \frac{A}{k} = \frac{T}{2\sqrt{y_0}}$$

Sub this into *

$$t = -\frac{2T}{2\sqrt{y_0}} \sqrt{y} + T$$

$$\frac{t}{T} = 1 - \frac{\sqrt{y}}{\sqrt{y_0}}$$

$$\sqrt{\frac{y}{y_0}} = 1 - \frac{t}{T}$$

$$\frac{y}{y_0} = \left(1 - \frac{t}{T}\right)^2$$

$$\underline{\underline{y = y_0 \left(1 - \frac{t}{T}\right)^2}}$$

iii) $t=10$ when $y=y_0/2$, $\therefore \frac{1}{2} = \left(1 - \frac{10}{T}\right)^2 \Rightarrow 1 - \frac{10}{T} = \frac{1}{\sqrt{2}}$

$$\Rightarrow \frac{10}{T} = 1 - \frac{1}{\sqrt{2}} = \frac{\sqrt{2}-1}{\sqrt{2}} \Rightarrow T = \frac{10\sqrt{2}}{\sqrt{2}-1} \text{ sec} = 10\sqrt{2}(\sqrt{2}+1) \text{ sec.}$$
$$= \underline{\underline{20 + 10\sqrt{2} \text{ sec.}}}$$

Question 14 (cont)

iii) $(x+i)^5 + (x-i)^5 = 0$

$$\begin{aligned} & \left(x^5 + {}^5C_1 x^4(i) + {}^5C_2 x^3(i)^2 + {}^5C_3 x^2(i)^3 + {}^5C_4 x(i)^4 + i^5 \right) \\ & + \left(x^5 + {}^5C_1 x^4(-i) + {}^5C_2 x^3(-i)^2 + {}^5C_3 x^2(-i)^3 + {}^5C_4 x(-i)^4 + (-i)^5 \right) = 0 \end{aligned}$$

The imaginary terms cancel

$$2x^5 + 2{}^5C_2 x^3 i^2 + 2{}^5C_4 x i^4 = 0$$

But ${}^5C_2 = 10$, ${}^5C_4 = 5$, $i^2 = -1$. $\therefore x^5 - 10x^3 + 5x = 0$ is the equivalent equation

$$x(x^4 - 10x^2 + 5) = 0$$

But the 0 root is given by x factor.

$\therefore$ roots of $x^4 - 10x^2 + 5 = 0$ are $\pm \cot \frac{\pi}{10}$, $\pm \cot \frac{3\pi}{10}$

iv) Solve: $x^2 = \frac{10 \pm \sqrt{100 - 20}}{2} = 5 \pm 2\sqrt{5}$

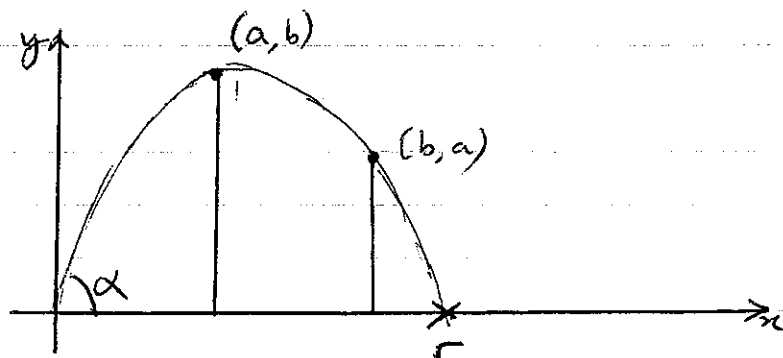
$\therefore$ Four roots are $\pm \sqrt{5 \pm 2\sqrt{5}}$.

But the positive roots are $\cot \frac{\pi}{10}$, $\cot \frac{3\pi}{10}$

The largest positive root is $\cot \frac{\pi}{10}$.

$$\therefore \cot \frac{\pi}{10} = \sqrt{5 + 2\sqrt{5}}$$

c)



The points (a,b) and (b,a) lie on the path

$$\begin{cases} \therefore b = a \tan \alpha - \frac{g a^2 \sec^2 \alpha}{2V^2} & (1) \\ a = b \tan \alpha - \frac{g b^2 \sec^2 \alpha}{2V^2} & (2) \end{cases} \quad \left. \begin{array}{l} \text{to be solved for} \\ V \text{ and } \alpha. \end{array} \right\}$$

Question 14

a) $\ddot{x} = -n^2 x$ ($2 = \frac{2\pi}{n} \rightarrow n = \pi$)

$$\therefore \frac{d}{dx} \left(\frac{v^2}{2} \right) = -\pi^2 x$$

$$\frac{v^2}{2} = -\frac{\pi^2 x^2}{2} + k$$

When $x = 3$, $v = 0$ $\therefore k = 9\pi^2/2$

$$\therefore v^2 = (9 - x^2)\pi^2$$

Max speed when $x = 0$ $|v|_{\max} = \underline{\underline{3\pi \text{ m/sec}}}$

Max acc. at $x = -3$ $\ddot{x}_{\max} = \underline{\underline{3\pi^2 \text{ m/sec}^2}}$

b) i) $(\cot \theta + i)^n + (\cot \theta - i)^n = \left(\frac{\cos \theta + i \sin \theta}{\sin \theta} \right)^n + \left(\frac{\cos \theta - i \sin \theta}{\sin \theta} \right)^n$
 $= \frac{1}{\sin^n \theta} \left\{ (\cos \theta + i \sin \theta)^n + (\cos(-\theta) + i \sin(-\theta))^n \right\}$
 (de Moivre) $= \frac{1}{\sin^n \theta} (\cos n\theta + i \sin n\theta + \cos(-n\theta) + i \sin(-n\theta))$
 $= \frac{1}{\sin^n \theta} (\cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta)$
 $= \underline{\underline{\frac{2 \cos n\theta}{\sin^n \theta}}}$

ii) Put $n=5$, $x = \cot \theta$ where θ are roots of $\frac{2 \cos 5\theta}{\sin^5 \theta} = 0$

i.e., $\cos 5\theta = 0$

$$5\theta = 2k\pi \pm \frac{\pi}{2} \quad (\text{general solution for } \cos)$$

$$= \pm \frac{\pi}{2} \quad (k=0),$$

$$2\pi \pm \frac{\pi}{2} \quad (k=1) \text{ or } -2\pi \pm \frac{\pi}{2} \quad (k=-1)$$

$$\therefore \theta = \pm \frac{\pi}{10}, \frac{5\pi}{10}, \frac{3\pi}{10}, -\frac{5\pi}{10}, -\frac{3\pi}{10}$$

The roots of equation are $\cot(\frac{\pi}{2})$, $\cot(\pm \frac{\pi}{10})$, $\cot(\pm \frac{3\pi}{10})$
 ($\cot(-\frac{\pi}{2})$ is same as $\cot(\frac{\pi}{2})$)

i.e. since $\cot(-\theta) = -\cot \theta$, roots are $\underline{\underline{0, \pm \cot \frac{\pi}{10}, \pm \cot \frac{3\pi}{10}}}$
 $\uparrow$
 $\cot(\frac{\pi}{2})$

Question 14 (cont)

$$\textcircled{1} \times b^2 \quad b^3 = ab^2 \tan \alpha - \frac{ga^2b^2 \sec^2 \alpha}{2V^2} \quad \textcircled{3}$$

$$\textcircled{2} \times a^2 \quad a^3 = a^2b \tan \alpha - \frac{ga^2b^2 \sec^2 \alpha}{2V^2} \quad \textcircled{4}$$

$$\textcircled{3} - \textcircled{4} \quad b^3 - a^3 = \tan \alpha \quad ab(b-a)$$

$$\therefore \tan \alpha = \frac{b^3 - a^3}{ab(b-a)} = \frac{b^2 + ab + a^2}{ab} \quad *$$

$$\textcircled{1} \times b \quad b^2 = ab \tan \alpha - \frac{ga^2b \sec^2 \alpha}{2V^2} \quad \textcircled{5}$$

$$\textcircled{2} \times a \quad a^2 = ab \tan \alpha - \frac{gab^2 \sec^2 \alpha}{2V^2} \quad \textcircled{6}$$

$$\textcircled{5} - \textcircled{6} \quad b^2 - a^2 = \frac{g \sec^2 \alpha}{2V^2} \cdot ab(b-a)$$

$$\frac{2V^2}{g \sec^2 \alpha} = \frac{ab}{a+b}$$

If the range is r , then $(r, 0)$ lies on the curve

$$\therefore 0 = r \tan \alpha - \frac{gr^2 \sec^2 \alpha}{2V^2}$$

Rejecting the $r=0$, solution

$$\frac{gr \sec^2 \alpha}{2V^2} = \tan \alpha$$

$$r = \tan \alpha \frac{2V^2}{g \sec^2 \alpha} = \frac{b^2 + ab + a^2}{ab} \frac{ab}{a+b} = \frac{b^2 + ab + a^2}{a+b}$$

$$\text{ii) From } * \quad \tan \alpha = \frac{b^2 + ab + a^2}{ab}$$

But since $(a-b)^2 > 0$ (since $b \neq a$, given)

$$a^2 + b^2 - 2ab > 0$$

$$a^2 + b^2 > 2ab$$

$$\therefore \tan \alpha > \frac{2ab + ab}{ab}$$

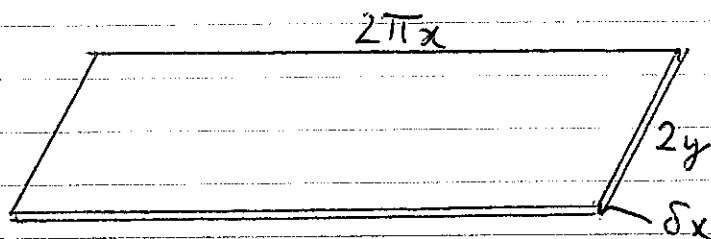
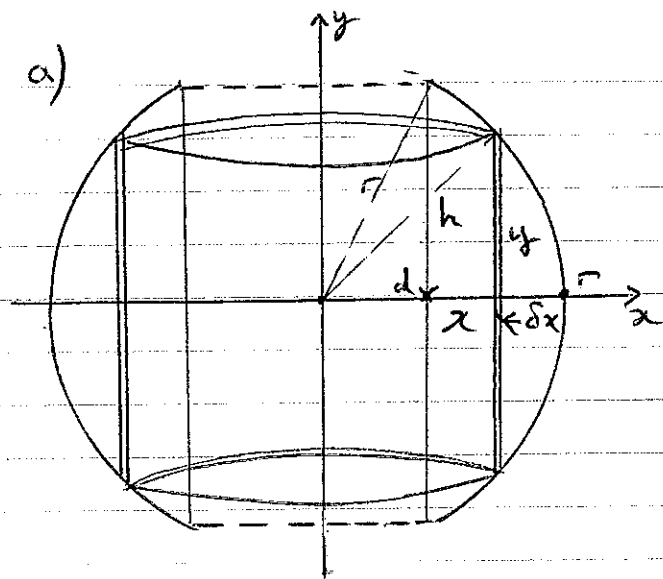
$$\tan \alpha > 3$$

$$\alpha > \tan^{-1}(3)$$

Since α is acute and $\tan^{-1}$ is an increasing function.

Question 15

a)



$$\delta V = 4\pi xy \delta x$$

$$\therefore V = \lim_{\delta x \rightarrow 0} \sum_{x=d}^r 4\pi xy \delta x$$

where d as in diagram

$$= 4\pi \int_d^r xy \, dx$$

But, by Pythagoras, $d = \sqrt{r^2 - h^2}$, $y = \sqrt{r^2 - x^2}$

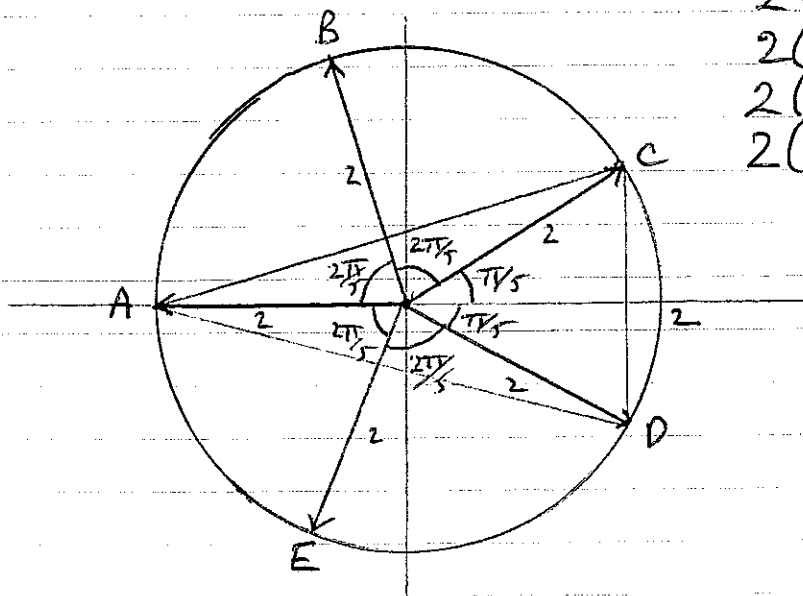
$$\therefore V = 4\pi \int_{\sqrt{r^2 - h^2}}^r x \sqrt{r^2 - x^2} \, dx$$

$$= \left[\frac{4\pi}{3} (-1) (r^2 - x^2)^{3/2} \right]_{\sqrt{r^2 - h^2}}^r$$

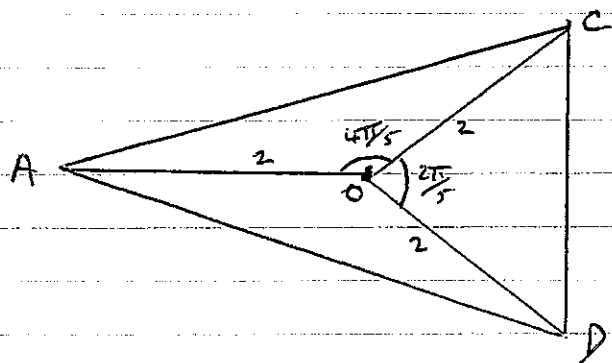
$$= \frac{4\pi}{3} \left\{ -0 + (r^2 - (r^2 - h^2))^{3/2} \right\} = \frac{4\pi h^3}{3} \quad (\text{independent of } r)$$

b) i) $z^5 = 2^5(-1)$ Roots are

$2(\cos \pi/5 + i \sin \pi/5)$	C
$2(\cos 3\pi/5 + i \sin 3\pi/5)$	B
$2(\cos \pi + i \sin \pi) = -2$	A
$2(\cos(-3\pi/5) + i \sin(-3\pi/5))$	E
$2(\cos(-\pi/5) + i \sin(-\pi/5))$	D



Question 15 (cont)



$$\begin{aligned}
 \text{Total area} &= \Delta COD + \Delta AOC + \Delta AOD \\
 &= \frac{1}{2} \times 2 \times 2 \sin \frac{2\pi}{5} + 2 \left(\frac{1}{2} \times 2 \times 2 \sin \frac{4\pi}{5} \right) \\
 &= 2 \sin \frac{2\pi}{5} + 4 \sin \frac{4\pi}{5} \\
 &= 1.902113 + 2.35114 \\
 &= \underline{\underline{4.25 \text{ u}^2 \text{ (to 2DP)}}}
 \end{aligned}$$

c)

Let M be the point (X, Y)

The chord of contact from M to the ellipse is given by

$$xX + 4yY = 4 \quad (1)$$

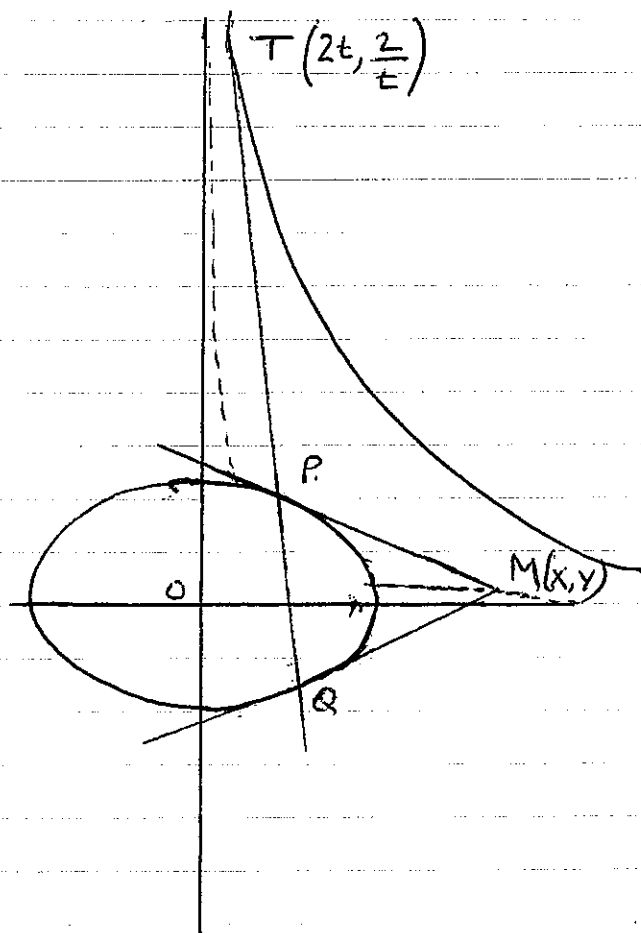
The tangent from T has equation

$$x \left(\frac{2}{t} \right) + 2ty = 8$$

$$\text{or } \frac{x}{t} + ty = 4 \quad (2)$$

① and ② are the same line

$$\left. \begin{aligned} X &= 1/t \\ Y &= t/4 \end{aligned} \right\} \text{ Eliminate } t \Rightarrow XY = \frac{1}{4} \quad \text{ locus of M is } \underline{\underline{xy = \frac{1}{4}}}$$



ii) This is a rectangular hyperbola, centred on origin. It will be "tighter" to the axes.

Chords of contact cannot be drawn from inside ellipse.

The curve crosses ellipse when $x^2 + \frac{4}{16x^2} = 4$

$$\left. \begin{aligned} \text{Restrictions on } x: & \quad x > \sqrt{2 + \frac{\sqrt{15}}{2}} \\ & \quad 0 < x < \sqrt{2 - \frac{\sqrt{15}}{2}} \end{aligned} \right\}$$

And similar on negative side.

$$\begin{aligned} 4x^4 - 16x^2 + 1 &= 0 \\ x^2 &= \frac{16 \pm \sqrt{240}}{8} \\ &= 2 \pm \frac{\sqrt{15}}{2} \end{aligned}$$

Question 16

a) Alan picks first:

$$\text{Prob. of success} = \frac{1}{n} + \frac{(n-1)(n-2)}{n(n-1)(n-2)} \frac{1}{n} + \frac{(n-1)(n-2)(n-3)(n-4)}{n(n-1)(n-2)(n-3)(n-4)} \frac{1}{n} + \dots$$

$\uparrow \quad \quad \uparrow \quad \uparrow \quad \uparrow \quad \quad \quad \dots + \frac{(n-1)(n-2)}{n(n-1)} \frac{4}{5} \frac{3}{4} \frac{2}{3} \frac{1}{2} \frac{1}{n}$

1st pick A fails B fails A picks

Each term is $\frac{1}{n}$, and there are $(n+1)/2$ terms

$$\therefore \text{Prob} = \underline{\underline{\frac{n+1}{2n}}}$$

$$\text{ii) Prob} = \frac{2}{n} + \frac{n-2}{n} \frac{n-3}{n-1} \frac{2}{n-2} + \dots + \frac{(n-2)(n-3)}{n(n-1)} \frac{4}{6} \frac{3}{5} \frac{2}{4} \frac{1}{3} \frac{2}{2}$$

$$= \frac{2}{n} \left(\frac{n-1}{n-1} + \frac{n-3}{n-1} + \frac{n-5}{n-1} + \dots + \frac{1}{n-1} \right)$$

$$= \frac{2}{n(n-1)} (1+3+5+\dots+(n-1))$$

$\uparrow$ A.P. with $a=1, d=2, "n" = \frac{n}{2}$

$$= \frac{2}{n(n-1)} \frac{n}{4} \left(2 + \left(\frac{n}{2} - 1 \right) 2 \right) = \frac{2}{n(n-1)} \frac{n^2}{4}$$

$$= \underline{\underline{\frac{n}{2(n-1)}}}$$

$$\text{iii) Prob} = \frac{2}{n} + \frac{(n-2)(n-3)}{n(n-1)(n-2)} \frac{2}{n-3} + \dots + \frac{(n-2)(n-3)}{n(n-1)} \frac{5}{7} \frac{4}{6} \frac{3}{5} \frac{2}{4} \frac{2}{3}$$

$$= \frac{2}{n} \left(\frac{n-1}{n-1} + \frac{n-3}{n-1} + \dots + \frac{2}{n-1} \right)$$

$$= \frac{2}{n(n-1)} (2+4+6+\dots+(n-1))$$

$\uparrow$ AP $a=2, d=2$ " n " = $\frac{n-1}{2}$

$$= \frac{2}{n(n-1)} \frac{(n-1)}{4} \left(4 + \left(\frac{(n-1)}{2} - 1 \right) 2 \right)$$

$$= \frac{2}{n(n-1)} \frac{(n-1)(n+1)}{4}$$

$$= \underline{\underline{\frac{n+1}{2n}}}$$

Question 16 (cont)

b) i) Solve $y = mx \pm \sqrt{a^2 m^2 + b^2}$ against $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Sub for y to get quadratic in x .

$$\frac{x^2}{a^2} + \frac{(mx \pm \sqrt{a^2 m^2 + b^2})^2}{b^2} = 1$$

$$\frac{x^2}{a^2} + \frac{m^2 x^2 + a^2 m^2 + b^2 \pm 2mx\sqrt{a^2 m^2 + b^2}}{b^2} - 1 = 0$$

$$b^2 x^2 + a^2 m^2 x^2 + a^4 m^2 + a^2 b^2 \pm 2a^2 m \sqrt{a^2 m^2 + b^2} x - a^2 b^2 = 0$$

$$x^2(a^2 m^2 + b^2) \pm 2a^2 m \sqrt{a^2 m^2 + b^2} x + a^4 m^2 = 0$$

$$(x\sqrt{a^2 m^2 + b^2} \pm a^2 m)^2 = 0$$

As this always gives a double root, the lines $y = mx \pm \sqrt{a^2 m^2 + b^2}$ are always tangents to the ellipse.

ii) Let $P(x, y)$ be external to the ellipse. Then the gradients of the 2 tangents from P are given by

$$Y = mX \pm \sqrt{a^2 m^2 + b^2}$$

$$Y - mX = \sqrt{a^2 m^2 + b^2}$$

$$Y^2 + m^2 X^2 - 2mXY = a^2 m^2 + b^2$$

$$m^2(X^2 - a^2) - 2mXY + (Y^2 - b^2) = 0$$

If these lines are perpendicular, the two roots m_1, m_2 must give $m_1 m_2 = -1$. This is the product of the roots.

$$\therefore \frac{Y^2 - b^2}{X^2 - a^2} = -1$$

$$Y^2 - b^2 = -X^2 + a^2$$

$$\underline{X^2 + Y^2 = a^2 + b^2}$$

Thus the locus of points where the two tangents are perpendicular is $x^2 + y^2 = a^2 + b^2$, a circle.

Question 16 (cont)

b) ii)

① AC is line $y = mx + \sqrt{a^2 m^2 + b^2}$

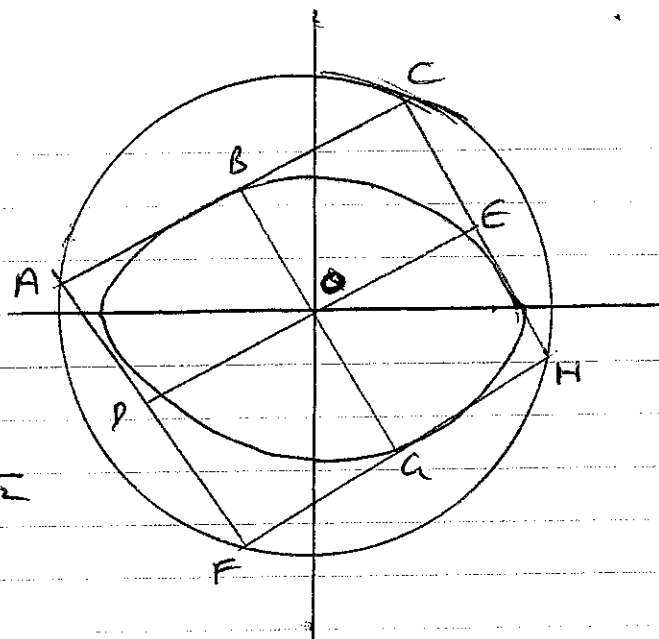
② FH is line $y = mx - \sqrt{a^2 m^2 + b^2}$

③ DE is line $y = mx$

④ CH is line $y = -\frac{x}{m} + \sqrt{\frac{a^2}{m^2} + b^2}$

⑤ AF is line $y = -\frac{x}{m} - \sqrt{\frac{a^2}{m^2} + b^2}$

⑥ BG is line $y = -\frac{x}{m}$



To find length CA, solve ① against the circle.

$$x^2 + (mx + \sqrt{a^2 m^2 + b^2})^2 = a^2 + b^2$$

$$x^2(1+m^2) + 2mx\sqrt{a^2 m^2 + b^2} + a^2 m^2 - a^2 = 0$$

If roots are x_1, x_2 then $(x_1 - x_2)^2 = (x_1 + x_2)^2 - 4x_1 x_2$

$$= \frac{4m^2(a^2 m^2 + b^2)}{(1+m^2)^2} - \frac{4a^2(m^2 - 1)}{(1+m^2)}$$

$$= \frac{4m^2(a^2 m^2 + b^2) - 4a^2(m^4 - 1)}{(1+m^2)^2}$$

$$= \frac{4(m^2 b^2 + a^2)}{(1+m^2)^2}$$

$$(y_1 - y_2) = m(x_1 - x_2) \quad \therefore (y_1 - y_2)^2 = \frac{4m^2(m^2 b^2 + a^2)}{(1+m^2)^2}$$

$$\therefore (CA)^2 \text{ is } \delta (\text{length})^2 (x_1 - x_2)^2 + (y_1 - y_2)^2 = \frac{4(m^2 b^2 + a^2)(1+m^2)}{(1+m^2)^2}$$

$$\therefore CA = \sqrt{\frac{4(m^2 b^2 + a^2)}{(1+m^2)}}$$

Similarly, replacing m with $-1/m$, $CH = \sqrt{\frac{4(b^2/m^2 + a^2)}{(1+1/m^2)}}$

$$= \sqrt{\frac{4(a^2 m^2 + b^2)}{(1+m^2)}}$$

$$\therefore \text{Area is } CA \times CH = \sqrt{\frac{4(a^2+m^2b^2)}{1+m^2}} \sqrt{\frac{4(a^2m^2+b^2)}{1+m^2}}$$

$$= \frac{4\sqrt{(a^2+m^2b^2)(a^2m^2+b^2)}}{1+m^2}$$

iv)

$$(a-b)^2 \geq 0$$

$$a^2+b^2-2ab \geq 0$$

$$a^2+b^2 \geq 2ab$$

Substitute $a=\sqrt{m}$, $b=\frac{1}{\sqrt{m}}$ (Requires $m>0$)

$$\therefore m + \frac{1}{m} \geq 2\frac{\sqrt{m}}{\sqrt{m}}, \text{ i.e., } \underline{m + \frac{1}{m} \geq 2 \text{ for } m > 0.}$$

v)

$$A^2 = 16a^2b^2 + \frac{16(a^2-b^2)^2}{(m+\frac{1}{m})^2}$$

Smallest A will be as $m+\frac{1}{m} \rightarrow \infty$, no contribution from second term

$$A_{\min}^2 = 16a^2b^2 \quad \underline{A_{\min} = 4ab}$$

Largest A will be when $m+\frac{1}{m} = 2$.

$$A_{\max}^2 = 16a^2b^2 + \frac{16(a^2-b^2)^2}{4} = 16a^2b^2 + 4(a^4+b^4-2a^2b^2)$$

$$= 8a^2b^2 + 4a^4 + 4b^4$$

$$= 4(a^4+b^4+2a^2b^2)$$

$$= 4(a^2+b^2)^2$$

$$\therefore A_{\max} = \underline{2(a^2+b^2)}$$